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Studierichting: Informatica

Oefeningenreeks 6E nr. 1.4

$$T_6(f, 1)(x) \text{ voor } f(x) = x^6 + x^3 - x^2 + 4$$

$$T_n(f, a)(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

$$T_6(f, 1)(x) = \sum_{k=0}^6 \frac{f^{(k)}(1)}{k!} (x-1)^k$$

$$f'(x) = (x^6 + x^3 - x^2 + 4)' = 6x^5 + 3x^2 - 2x$$

$$\Rightarrow f'(1) = 7$$

$$f''(x) = (x^6 + x^3 - x^2 + 4)'' = 30x^4 + 6x - 2$$

$$\Rightarrow f''(1) = 34$$

$$f'''(x) = (x^6 + x^3 - x^2 + 4)''' = 120x^3 + 6$$

$$\Rightarrow f'''(1) = 126$$

$$f^{iv}(x) = (x^6 + x^3 - x^2 + 4)^{iv} = 360x^2$$

$$\Rightarrow f^{iv}(1) = 360$$

$$f^v(x) = (x^6 + x^3 - x^2 + 4)^v = 720x$$

$$\Rightarrow f^v(1) = 720$$

$$f^{vi}(x) = (x^6 + x^3 - x^2 + 4)^{vi} = 720$$

$$\Rightarrow f^{vi}(1) = 720$$

$$\begin{aligned} T_6(f, 1)(x) &= \frac{f(1)}{0!} (x-1)^0 + \frac{f'(1)}{1!} (x-1)^1 + \frac{f''(1)}{2!} (x-1)^2 + \frac{f'''(1)}{3!} (x-1)^3 + \\ &\frac{f^{iv}(1)}{4!} (x-1)^4 + \frac{f^v(1)}{5!} (x-1)^5 + \frac{f^{vi}(1)}{6!} (x-1)^6 \\ &= 5 + 7(x-1) + 17(x-1)^2 + 21(x-1)^3 + 15(x-1)^4 + 6(x-1)^5 + (x-1)^6 \end{aligned}$$

References